C Operators: Precedence & Associativity

Precedence: Precedence determines the **priority** of operators in an expression. An operator with higher precedence is evaluated before an operator with lower precedence.

Associativity: Associativity determines the **direction of evaluation** when operators have the same precedence. It can be **Left → Right** or **Right → Left**.

Precedence	Operator(s)	Description	Associativity
1 (Highest)	()	Brackets / Parentheses	Left → Right
2	++ --	Post-increment, Post-decrement	Left → Right
3	Pre-increment (++) Pre-decrement (--)	Pre-increment and Pre-decrement	Right → Left
4	* / %	Multiplication, Division, Modulus	Left → Right
5	+ -	Addition, Subtraction	Left → Right
6	<< >>	Bitwise Left/Right Shift	Left → Right
7	< <= > >=	Relational operators	Left → Right
8	== !=	Equality operators	Left → Right
9	&	Bitwise AND	Left → Right
10	^	Bitwise XOR	Left → Right
11	|	Bitwise OR	Left → Right
12	&&	Logical AND	Left → Right
13	||	Logical OR	Left → Right
14	?:	Conditional / Ternary operator	Right → Left
15	= += -= *= /= %=	Assignment operators	Right → Left

Quick Rule: Precedence = **Priority** | Associativity = **Direction** | Parentheses **()** can explicitly control evaluation order.